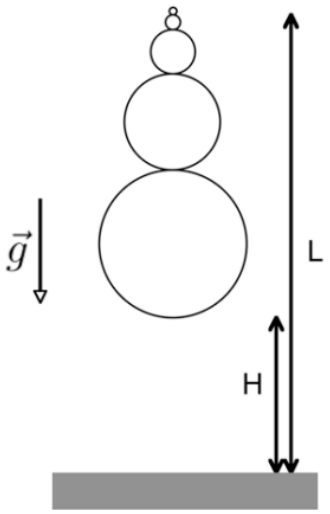
Problem 9 - The super-astro-blaster



9. **The super-astro-blaster.** [15 points]

In class we demonstrated how the astroblaster (a set of elastic bouncing balls released from rest and rebounding from the floor) can launch the smallest and topmost ball to considerable distance. Let's push this to the test. Assume we have n balls of mass (left figure above) m_1, m_2, m_3, ..., m_n where $m_1 >> m_2 >> m_3 >> ... > m_n$. The most massive ball is at bottom and starts at rest at a height H = 1 meter above the ground. Assume that all balls bounce elastically, that constant gravitational acceleration $\vec{g}$ points downward, and that air resistance and realistic limits in the elasticity of the balls can be ignored.

(a) (10 points) In terms of n, to what height does the top ball bounce? How many balls do you need to reach 1 kilometer?

(b) (5 points) How many balls do you need to achieve escape velocity? You will need to determine an expression for escape velocity from the Earth, which is possible (to within an order of magnitude) through dimensional analysis.

Solution (handwritten) follows on the next pages.

camlin
Date :

Problem 9.

Assumptions :- 1) Balls collide in pairs.
2) In an elastic collision, the ball with the lesser weight, Bounces of with equal and opposite relative velocity.
The ball with the higher weight maintains its

1st collision

$$V_{ball\,1} = -V_i = -\sqrt{2gh}$$

$(n+1)^{th}$ ball collides with the nth ball.

Assume the nth ball is moving up with some velocity V_n (Ground Frame). and the $(n+1)^{th}$ ball is moving downward with V_i $(-\sqrt{2gh})$

Relative velocity of the $(n+1)^{th}$ ball from the frame of the nth ball :-

$$V_{rn+1} = -(V_i + V_n)$$

After the collision.

$$V_{rn+1} = (V_i + V_n)$$

From the Ground Frame

$$V_{n+1} = 2V_n + V_i$$

Homogeneous Sol'n

$\alpha = 2 \rightarrow v_n = C_1 2^n$

Particular Sol'n Guess $v_n = c$

$c = 2c + v_i$

$c = -v_i$

General Sol'n

$v_n = C_1 (2^n) - v_i$

Boundary Conditions $v_1 = v_i$

$C_1 = v_i$

$v_n = v_i (2^n - 1)$

$v_n = \sqrt{2gh} (2^n - 1)$

The top ball collides at $h = L$

Max height $= \frac{2gh (2^n - 1)^2}{2g} = h(4^n + 1 - 2^{n+1})$

6) $v_n = \sqrt{2gh}\,(2^n - 1)$

$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2Rg}$

$\sqrt{2gh}\,(2^n - 1) = \sqrt{2Rg}$

$\sqrt{h}\,(2^n - 1) = \sqrt{R}$

$(2^n) = 1 + \sqrt{\frac{R}{h}}$

$n = \log_2\left(1 + \sqrt{\frac{R}{h}}\right)$